

Ten Tools

for Configuration Management



Configuration management (CM) comprises the detailed description and updating of information related to an enterprise's software and hardware. This article lists the top 10 free and open source tools that can help you carry out configuration management with ease.

Configuration management (CM) usually includes the updates and the versions that have been applied to software packages, as well as locations and network addresses of hardware devices. When a system needs a software or hardware upgrade, the user can access the CM program and database to see what is currently installed and then make a more informed decision about the upgradation needed.

The advantages of software configuration management (SCM) are:

- It reduces redundant work
- It effectively manages simultaneous updates
- It avoids configuration related problems
- It simplifies coordination between team members
- It is helpful in tracking defects

Top ten tools for configuration management

Chef

Chef is one of the most popular SCM tools. It is basically a framework for infrastructure development. It provides support and packages for framing one's infrastructure as code. It offers libraries for building up an infrastructure,

which can be deployed easily. It produces consistent, shareable and reusable components, which are known as recipes and are used to automate infrastructure. It comprises the Chef server, workstation, repository and the Chef client.

Chef server: This is the central registry or brain of the entire process and acts as a hub. It carries information about the infrastructure and the cookbooks, which are used to instruct Chef on how each node must be configured.

Chef client: This is installed on the nodes; it contacts the Chef server, fetches cookbooks and executes them on the nodes. The chef-client does the actual configuration.

Chef provides various tools for designing the infrastructure. These are based on high-level programming languages such as Python and Ruby. Some of Chef's tools are:

- Ohai, a system profiling tool
- Shuf, an interactive debugging console for testing and exploring Chef's behaviour
- Chef-Solo, a configuration management tool
- Chef client, an agent that runs on the system managed by Chef to communicate with the Chef server
- Knife, a multi-purpose command line tool